

Project Report

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1 Report

This project contains a didactic webpage explaining, and illustrating using images, Theorem 1 and Lemma 1/Theorem 3 and their proofs from [1]. Additionally, we describe the algorithms used to find inscribed polygons with minimum area or perimeter, and offer the user interactive canvases allowing them to run the algorithms on polygons of their choosing.

2 Use of AI

AI was not used to generate any code or text used in this project.

References

- [1] Ködmön, C. L., Lángi, Z. (2021). *Extremal convex polygons inscribed in a given convex polygon*, Computational Geometry, 102, 101844.
- [2] Deniz, A., Ratiu, A. (2009). *On the existence of Fagnano trajectories in convex polygonal billiards*, Regul. Chaot. Dyn. 14, 312-322.
- [3] Besant, W. H. (1895). *Conic Sections Treated Geometrically*, G. Bell.
- [4] Ausserhofer, M., Dann, S., Lángi, Z., Tóth, G. (2019). *An algorithm to find maximum area polygons circumscribed about a convex polygon*, Discrete applied mathematics, 255, 98-108.

(Another copy of this bibliography is provided on the webpage)